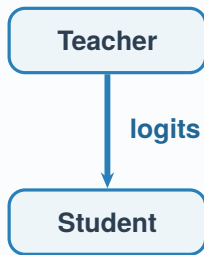


Pure KD

Distillation only



$y \sim p_{\text{data}}$

GKD

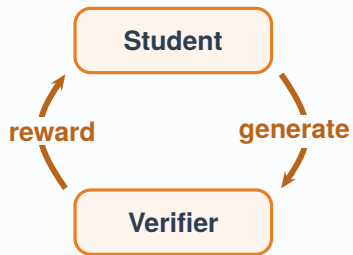
DistiLLM

MiniLLM

+ reward

Pure RL

Reward optimization



$y \sim p_{\theta}$

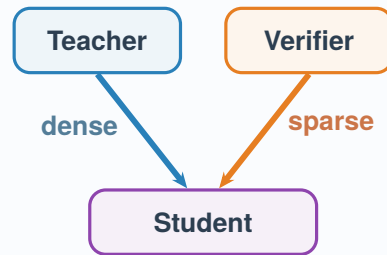
GRPO

PPO

unify

Hybrid KD+RL

Best of both worlds



$y \sim p_{\theta}$

RLKD

G-OPD

ReinfKD

$$\max_{\theta} \underbrace{\mathbb{E}_{y \sim p_{\theta}}[R(y)]}_{\text{RL reward}} - \beta \cdot \underbrace{D_{\text{KL}}(p_{\theta} \| p_T)}_{\text{KD matching}}$$